Gautham Ramkumar

Robot Control & Manipulation Engineer

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F-1 STEM OPT eligible; no immediate sponsorship required

Summary

Robotics M.S. candidate working across control synthesis and robot autonomy, from Hybrid Zero Dynamics gait design and feedback linearization to language-driven manipulation running fully on-device.

Education

Northeastern University, Boston, MA **2024 – 2026**
M.S. Robotics — Concentration in Electrical & Computer Engineering

- **Coursework:** Advanced Perception, Robot Sensing & Navigation, Autonomous Field Robotics, Mobile Robotics, Robot Mechanics & Controls, Legged Robotics

Vellore Institute of Technology (VIT), Chennai, India **2020 – 2024**
B.Tech. Electronics & Communications Engineering

Technical Skills

Languages: Python, C++, MATLAB
Control & Dynamics: Hybrid Zero Dynamics, Feedback Linearization, Trajectory Optimization, Lagrangian Dynamics, PID / LQR / MPC, Legged Locomotion
State Estimation & Calibration: Extended Kalman Filters, Factor Graph Optimization, Bundle Adjustment, Multi-Sensor Calibration, Camera-LiDAR Calibration, Temporal Synchronization, Sensor Fusion, Pose Estimation
Frameworks & Tools: ROS2, PyTorch, OpenCV, GTSAM, Open3D, TensorRT, CUDA, COLMAP, Docker, Git
Hardware & Sensors: ZED Mini Stereo Camera, Intel RealSense D435, LiDAR, VectorNav IMU, NVIDIA Jetson Nano

Selected Projects

Hybrid Zero Dynamics Bipedal Gait Control — 1.73 m/s walking, 1.1584 MCOT **2026**
MATLAB · Feedback Linearization · fmincon

- Derived full swing-phase Lagrangian dynamics and a heel-strike impact map symbolically for a 3-link planar biped.
- Imposed 4th-order Bézier virtual constraints to reduce the system onto a 2-DOF Hybrid Zero Dynamics manifold, then optimized the periodic orbit for minimum control effort with fmincon.

YesChef — Voice-Controlled Robotic Manipulation — 2nd place, Revolute Hackathon **Aug 2026**
Whisper · Qwen 2.5 VLM · Qwen 2.5 LM

- Built a fully local voice-to-action pipeline driving robotic manipulation for cooking tasks.
- Chained Whisper for speech-to-text, a Qwen 2.5 3B vision-language model for workspace understanding, and a Qwen 2.5 3B language model for instruction parsing and action planning.

Indoor 3D Mapping with RTAB-Map — 90% ground-truth alignment, GPS-denied **Sep 2025 – Oct 2025**
ROS2 · ZED Mini · LiDAR · VectorNav IMU

- Deployed RTAB-Map visual SLAM on a ZED Mini stereo camera in ROS2, bringing up a full sensor suite (LiDAR, VectorNav IMU) with calibrated coordinate transforms.
- Validated loop-closure detection and pose-graph consistency across long trajectories in repetitive, GPS-denied tunnels.

Camera-LiDAR Temporal Calibration — -19.49 ms offset, stable to 1.5 ms **Oct 2025 – Dec 2025**
Python · C++ · GTSAM · ROS2 · Open3D · OpenCV

- Built a stamp-primary temporal synchronization pipeline anchored on hardware capture timestamps, with a bounded residual path correcting latency bias via cross-modal edge alignment.
- Estimated a stable -19.49 ms camera-LiDAR offset across 1,500 camera frames and 750 LiDAR scans, repeatable to within 1.5 ms across windows.

Experience

SmartInternz **2023**
IoT Engineering Intern

- Built real-time sensing pipelines and cloud logging systems on IoT hardware.
- Integrated embedded sensor data acquisition with cloud infrastructure.

Awards

2nd Place, Revolute Hackathon (Aug 2026) — Voice-controlled robotic manipulation for cooking, built in a weekend.